

Aesha Gandhi

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EDUCATION

Duke University

Durham, NC

Master's in Data Science GPA: 3.962

Aug. 2025 – May 2027 (Expected)

- Coursework: Agentic AI, Deep Learning, Reinforcement Learning, Statistics, Data Engineering, NLP, Machine Learning

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Data Science GPA: 3.72

Aug. 2021 – May 2025

- Domain emphasis in Applied Mathematics and Modeling; Minors in Computer Science, Industrial Engineering
- Coursework: Artificial Intelligence, Algorithms, Linear Programming, Inference/Decisions, Data Mining, Discrete Mathematics, Probability Theory, Data Ethics, Data Structures, Database Systems

North Carolina School of Science and Math (NCSSM)

Durham, NC

EXPERIENCE

SAS

Cary, NC

Applied Artificial Intelligence and Modeling Intern (Year-Round)

May 2025 – Present

- Developing a real-time computer vision pipeline in Python for forklift safety detection in warehouse environments, fusing YOLO-based object detection, instance segmentation, and keypoint estimation with ByteTrack multi-object tracking to assess clamp-roll alignment via geometric feature computation and temporal aggregation, triggering alerts on unsafe handling events
- Built production ready Python and TypeScript SDKs for containerized ML model deployment and management, leveraging OpenAPI based code generation, testing, packaging, and CI/CD integration
- Developed the analytics foundation for a centralized engineering metrics platform by querying large-scale DevOps data with SQL and Python, defining engineering KPIs, and building an automated dependency intelligence pipeline to collect, ingest, and report Python and RPM dependencies across deployable units.

Interactive Analytics and Modeling Intern (Year-Round)

May 2022 – May 2025

- Designed, implemented, and shipped a hyperparameter tuning results feature for SAS Viya Visual Analytics used by 700+ enterprise customers, building React based front end dialogs and visualizations that surfaced training progress, objective history, and model performance across tree based models, SVMs, and neural networks

College of Computing, Data Science, and Society

Berkeley, CA

Course Staff Tutor

Jan. 2024 – May 2024

- Graded 250+ students' work and led weekly office hours for a course in probabilistic inference and decision making

PROJECTS

Intelligent Credit Card Advisor Agent | *LLMs, OCR, Transformer Classifiers, Streamlit, Python* Spring 2026

- Developed an agentic receipt to recommendation pipeline that extracts spending patterns from receipt images using CV and OCR (Tesseract, PaddleOCR), classifies line items with a transformer based classifier, and leverages LLM powered web search tool use to generate grounded, real time credit card recommendations

Codenames Spymaster RL Agent | *PyTorch, Sentence Transformers, Python* Spring 2026

- Designed a goal-conditioned RL agent for the Codenames spymaster task using sentence-transformer embeddings for semantic board representation, benchmarking Greedy, Behavioral Cloning, and Soft Actor-Critic with Hindsight Experience Replay pipelines to study imitation warm-starts and reward shaping for semantic clue generation

Vision-Language Image Captioning | *Hugging Face Transformers, PyTorch* Fall 2025

- Reversed the text-to-image generation paradigm by fine-tuning vision-language models (ViT-GPT2, BLIP) on 10,000 DiffusionDB image-caption pairs, achieving a 102% improvement in combined BLEU/BERTScore on NVIDIA A100 GPUs

AI Health Companion | *LLMs, AI Agents, Real-Time Systems* Hackathon Project

- Built a voice-first AI agent that uses real-time LLM audio interaction and multi-stage prompt pipelines to extract structured health signals from natural conversations with senior adults and surface early, severity-based wellbeing insights for families via a dashboard

Sparkify Music Recommendation Pipeline | *Apache Airflow, Postgres, Python, Streamlit* Fall 2025

- Built an end-to-end data engineering and recommendation system that simulates a production music platform by ingesting track metadata and listening events, orchestrating ETL, embedding generation, and scheduled inference, and serving personalized recommendations through an interactive dashboard

TECHNICAL SKILLS

Languages: Python, Java, R, JavaScript/TypeScript, SQL, SAS Base
Frameworks: React, PyTorch, Scikit-learn, Transformers, LLMs, TensorFlow, Apache Spark/PySpark, Apache Airflow/Kafka
Developer Tools: Git/GitHub, Docker, Databricks, AWS/GCP/Azure, Postgres, MongoDB
Libraries: Pandas/Polars, NumPy, Matplotlib